

WORKED SOLUTIONS

# ENDY2.2 MOTION: NON-CONSTANT ACCELERATION

**Question 1** (Hibbeler, R.C, 2010, Engineering Mechanics: Dynamics 12<sup>th</sup> Ed. Pearson)

The position of a particle is given by  $s = t^3 - 3t^2$  m, where  $t$  is in seconds. Calculate the time(s) when (a) the velocity of the particle is zero, (b) the acceleration is zero, and (c) calculate the total distance travelled by the particle after 3 seconds.

**Worked Solution 1** (a)  $s = t^3 - 3t^2 \Rightarrow v = \frac{ds}{dt} = 3t^2 - 6t$

when  $v = 0$ ,  $3t^2 - 6t = 0 \Rightarrow 3t(t-2) = 0$   
 $\Rightarrow \underline{t = 0, 2s}$

(b)  $v = 3t^2 - 6t \Rightarrow a = \frac{dv}{dt} = 6t - 6$

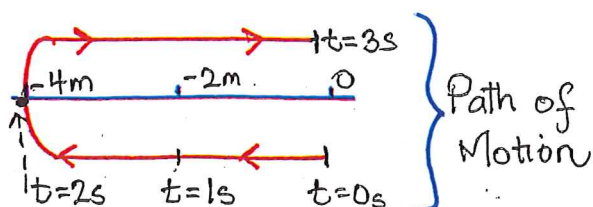
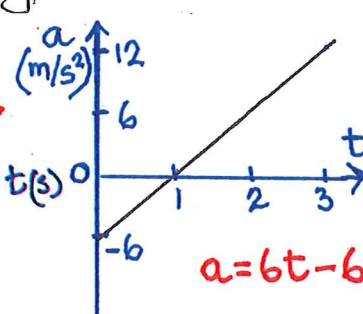
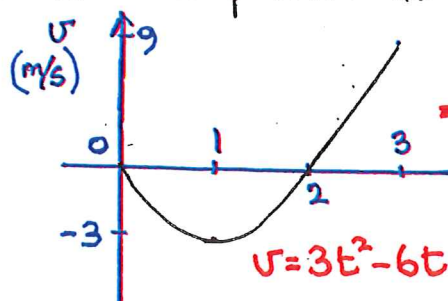
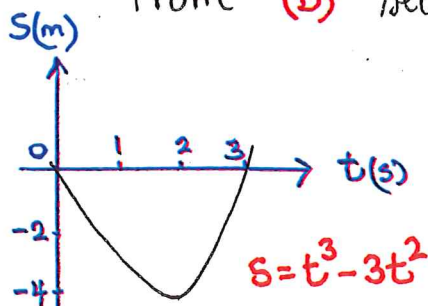
when  $a = 0$ ,  $6t - 6 = 0 \Rightarrow \underline{t = 1s}$

(c)  $s = t^3 - 3t^2 = t^2(t-3)$  Sketch a graph to see what's happening.

when  $s = 0$ ,  $t = 0, 3s$  ( $s$ -intercepts on  $s$ - $t$  graph)

From (a)  $t = 2s$  is turning point  $0 \leq t \leq 3$

From (b) second derivative is positive  $\therefore$  negative T.P.



Distance travelled

|                   |                     |
|-------------------|---------------------|
| $0 \leq t \leq 2$ | $d = 4m$ to 'left'  |
| $2 < t \leq 3$    | $d = 2m$ to 'right' |

Total Distance travelled = 6m

**Question 2** (Hibbeler, R.C, 2010, Engineering Mechanics: Dynamics 12<sup>th</sup> Ed. Pearson)

A particle is moving along a straight line such that its acceleration is given by  $a = (-2v) \text{ m/s}^2$ . If  $v = 20 \text{ m/s}$  when  $s = 0$  and  $t = 0$ , calculate (a) the particle's velocity (b) the particle's position and (c) the particle's acceleration when  $t = 2$  seconds.

**Worked Solution 2**

$$(a) \quad a = \frac{dv}{dt} = -2v \quad \Rightarrow \quad \frac{1}{2} \int \frac{dv}{v} = - \int dt$$

$$\text{when } t=0, v=20 : \quad \frac{1}{2} \int_{20}^v \frac{dv}{v} = - \int_0^t dt$$

$$\frac{1}{2} [\ln v]_{20}^v = -t \quad \Rightarrow \quad \frac{1}{2} (\ln v - \ln 20) = -t$$

$$\ln\left(\frac{v}{20}\right) = -2t \quad \Rightarrow \quad \frac{v}{20} = e^{-2t} \Rightarrow v = 20e^{-2t}$$

$$\text{when } t=2, \quad v = 20e^{-4} = \underline{\underline{0.366 \text{ m/s}}}$$

$$(b) \quad v = \frac{ds}{dt} = 20e^{-2t} \quad (\text{see above})$$

$$\Rightarrow ds = 20e^{-2t} dt \quad \Rightarrow \quad \int ds = 20 \int e^{-2t} dt$$

$$\text{when } t=0, s=0 : \quad \int_0^s ds = 20 \int_0^t e^{-2t} dt$$

$$s = 20 \cdot -\frac{1}{2} [e^{-2t}]_0^t = -10(e^{-2t} - 1)$$

$$\text{when } t=2, \quad s = -10(e^{-4} - 1) = \underline{\underline{9.817 \text{ m}}}$$

$$(c) \quad \text{Since } a = -2v \quad \text{and} \quad v = 20e^{-2t}$$

$$a = -2(20e^{-2t}) = -40e^{-2t}$$

$$\text{when } t=2, \quad a = -40e^{-4} = \underline{\underline{-0.733 \text{ m/s}^2}}$$